Ken (Tongli) Su

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CS PhD student working on **LLM agents, alignment, interpretability, and safety**, especially multi-agent collaboration and theory of mind: agents that reason about collaborators' beliefs, adapt to human values, and know when to seek human input. Co-first author of **Software World Models** (predicting what a change breaks before it runs) and **LARGER** (beats 7 baselines incl. Claude Code, Codex); first author, EMBC 2026 oral. Reviewer, NeurIPS 2026.

Education

Emory University — Atlanta, GA
Ph.D. in Computer Science — Aug 2025 – Dec 2028 (expected)

Columbia University — New York, NY
M.S. in Data Science, GPA: 3.95/4.00 — Dec 2023

Emory University — Atlanta, GA
B.A. in Computer Science & B.A. in Economics (Dual Major), GPA: 3.84/4.00 — May 2022

Selected Publications & Research

[1] **T. Su**[*], Y. Hu[*], L. Zhao, B. Zhu, J. Somasundaram, H. Haque. "Software World Models: From Consequence Prediction to Decision Value." *arXiv:2610.04940*, 2026. *(Under review; [*]equal contribution.)*

- Formulated **system-level consequence prediction** for coding agents: predict the **blast set** — the downstream services, libraries, and datastores a locally correct change will break — *before* execution, beyond what the agent itself observes.
- Built **Explore–Learn–Act**: execute counterfactual changes from restored system states, train Qwen2.5-Coder with **GRPO** on executed outcomes (plus an R-GCN graph-token variant with **SC-GRPO**), and act on per-consumer break probabilities.
- Blast-set F1 **0.431** → **0.571** vs. static reachability and ranking regret **more than halved**; on held-out real libraries, agent return **0.453** → **0.613**, above the CWM and Qwen-AgentWorld world-model baselines.

[2] Y. Hu, **T. Su**, Y. Sun, S. Ma, Q. Tan, L. Zhao. "Adaptive Memory Index Evolution for Agents." *arXiv:2610.34977*, 2026. *(Under review.)*

- **AMIE**: a plug-in **memory index evolution** layer that distills an LLM teacher's keep/suppress judgments into a Qwen3-8B LoRA policy and accumulates them into persistent per-memory access state, with no change to stored content or host retraining. Improved downstream F1/accuracy in **all 15** comparisons (**5 memory systems** incl. Mem0, A-MEM × **3 benchmarks**) by up to **25%** relative.

[3] Y. Hu[*], **T. Su**[*], L. Zhao, B. Zhu, H. Haque. "LARGER: Lexically Anchored Repository Graph Exploration and Retrieval." *arXiv:2605.16352*, 2026. *(Under review; [*]equal contribution.)*

- Anchored a coding agent's own lexical search hits into a repository graph and expanded confidence-filtered neighborhoods *inside* its search loop (imports, call chains, code–test links) with no added agent actions; built **AST-based graph construction** end to end.
- **+13.9 Acc@5** and **+12.2 Recall@5** over the strongest of **7 baselines** (BM25, Agentless, LocAgent, SWE-agent, OpenHands, Claude Code, Codex) across **4 benchmarks** spanning localization, test generation, and codebase QA.

[4] **T. Su**, A. Rafiei, M. van Assen, R. Sameni, G. D. Clifford, F. Marzbanrad, N. Katebi. "Cross-Modal Generative Framework for Signal Translation from Fetal–Maternal Electrocardiograms to Fetal Doppler Waveforms." *IEEE EMBC 2026*. **Accepted — Oral Presentation.**

- Designed a cross-modal generator (dilated convolutions + **cross-modal attention** + self-attention) for ECG-to-Doppler translation: PSD MSE **51% below** baseline; ablations show naive concatenation adds nothing while cross-modal attention cuts error **39%**.

[5] **T. Su**, N. Barari, N. Katebi. "A Review of Multimodal Pretraining Strategies for Physiological Wearable Sensor Foundation Models." *Under review*, 2026. — Taxonomy of five multimodal pretraining paradigms over **22 papers** across PPG/ECG/EEG/ACC.

Industry Experience

Machine Learning Engineer, GenAI | DuPont Specialty Products USA, LLC — New York, NY | Jul 2024 – Aug 2025

- Built a lead identification pipeline on **Azure Databricks** (Spark, Delta Lake, generative models) behind an internal sales co-pilot that became the company's primary prospecting tool (**$60M+** estimated value); fine-tuned **Sentence-BERT** with triplet loss for retrieval and ranking: **Recall@5 0.86**, **MRR 0.64**, **+27%** over the incumbent GPT-based system with hallucinations eliminated.
- Engineered an async, load-balanced **OpenAI API** enrichment module: runtime 10 h → 6 h (**40%**), token costs **–$12K/yr**.

Machine Learning Engineer | ICBC Financial Service, LLC — New York, NY | Feb 2024 – May 2024

- Developed an in-house **transformer foundation model** with a Bing Search API continual re-pretraining loop; **few-shot** financial report translation reached **BLEU 0.59** (cumulative 4-gram), cutting translation costs **$30K/yr** and turnaround **50%**.

Machine Learning Scientist | Unilever United States, Inc — New York, NY | Sep 2023 – Dec 2023

- Built a **multi-modal neural network** (PyTorch/Keras) fusing tabular product features with fine-tuned VGG16 image embeddings, deployed on **GCP** for real-time forecasting: sales-volume $\mathbf{R^2}$ **+54%** over XGBoost, optimizing **$3M** in supply chain cost.

Technical Skills

Agents & LLMs	LLM & coding agents, world models, agent memory, RL post-training (GRPO), fine-tuning (LoRA, distillation), retrieval/ranking (RAG, BM25, graph retrieval), Hugging Face Transformers, Sentence-BERT, OpenAI API, agent evaluation
ML & Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost; graph neural networks (R-GCN), attention, cross-modal & generative models, self-supervised pretraining, time-series & biosignals
Languages & Infra	Python, SQL, R, Java, C++, Bash; Spark, PySpark, Pandas, Databricks, Airflow, PostgreSQL; AWS, Azure, GCP